

Statistics Unit 6: Multiple Contrasts

Lecture Notes

Example: Three treatments for depression

1. Lavender Buds Rose Petals & Eucalyptus Oil Therapy
2. Ammonia Clorox & Coffee Therapy
3. (placebo control) Drinking Glass Red Stapler & Wood Block Therapy

Two types of questions:

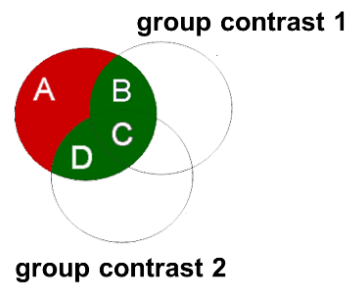
1. Focused questions, Dichotomous comparison: Is _____ different from _____ ?
2. A contrast between 2 means

Possible contrasts between three groups

	vs.		$\bar{y}_1 - \bar{y}_2 = ?$
	vs.		$\bar{y}_1 - \bar{y}_3 = ?$
	vs.		$\bar{y}_2 - \bar{y}_3 = ?$

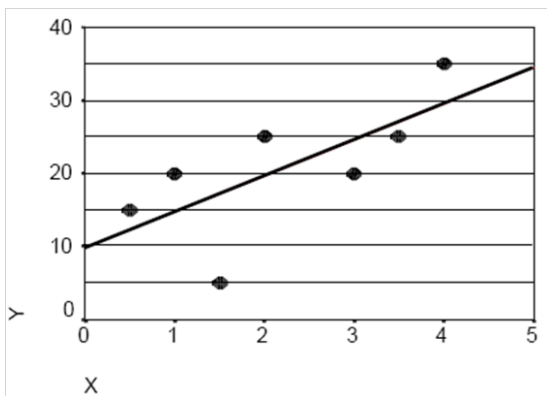
Omnibus Question

How much variance in Y
is explained by multiple group contrasts?



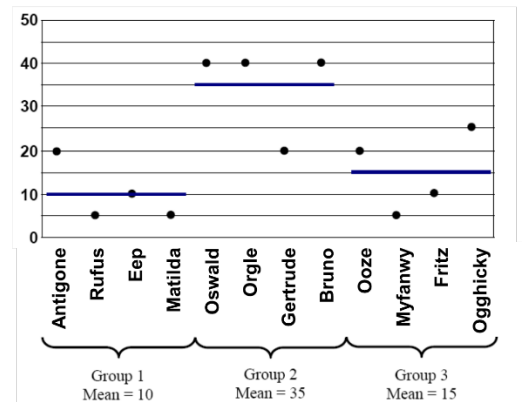
Regression

X is numerically scaled
Slope gives predicted scores for Y



Group contrasts

X is categorical
Group means give predicted scores for Y



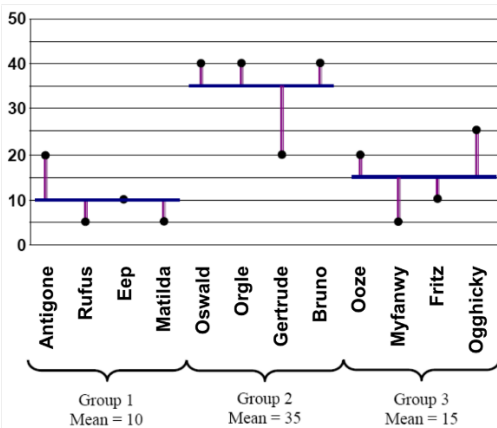
Sum of squares
 Explained variance
 Error variance
 Total variance

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained	Total – Error	p	SS / df	$\frac{MS\ Explained}{MS\ Error}$
Error		$n - p - 1$	SS / df	
Total		$n - 1$		

Error variance

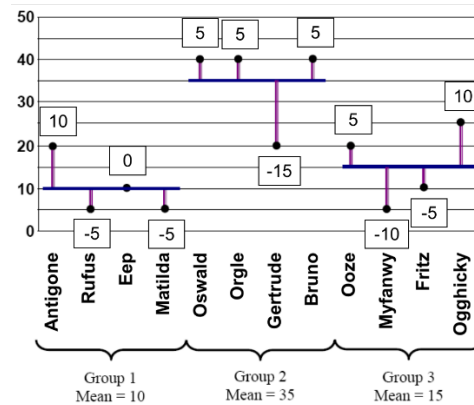
ANOVA:

Error is distance between observed score and group mean.



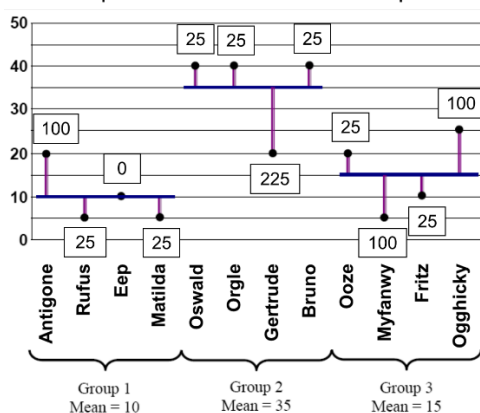
AKA
 within group
 variance
 residual
 variance

Calculating within group variance (error variance)



First compute
 the error for
 each person

Sum of Squared Error
 Square the errors. Add them up.



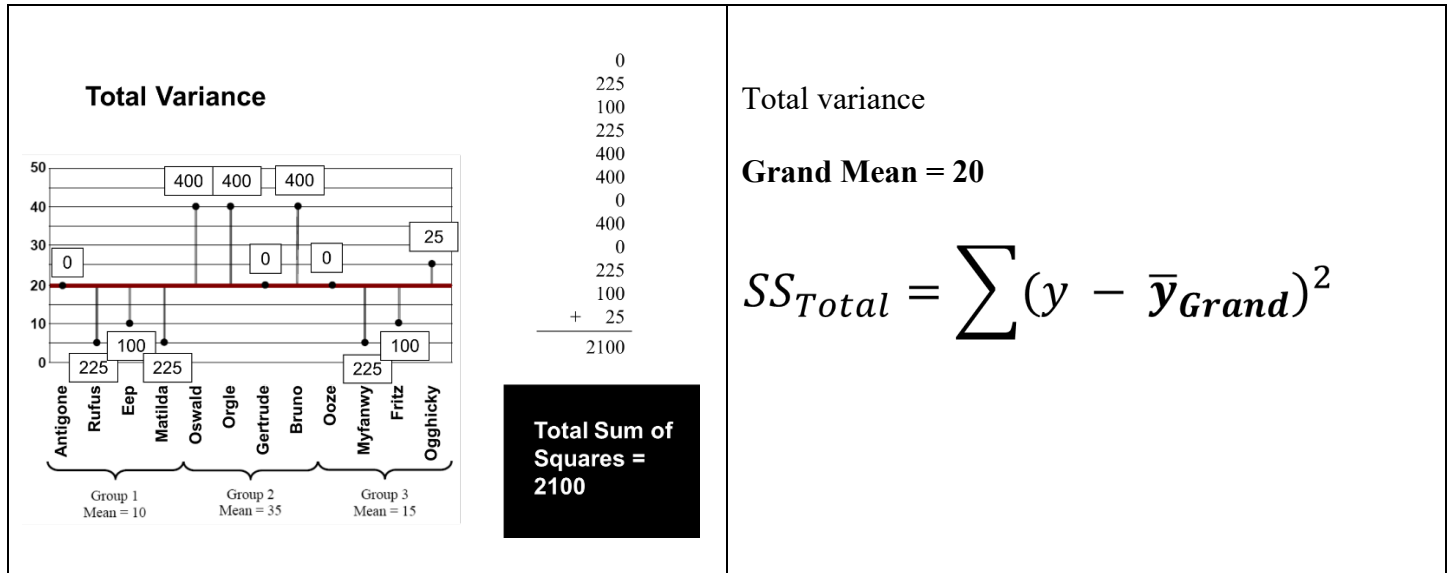
100
 25
 0
 25
 25
 25
 225
 25
 25
 100
 25
 100
 + 100
 700

Sum of squared
 error (within
 group sum of
 squares) = 700

Error variance

$$SS_{Error} = \sum (y - \bar{y}_{group})^2$$

Total variance



Explained variance

Option A:

Total sum of squares – Error sum of squares

Option B:

$\sum (\text{Squared GROUP Deviation}) (\text{Group N})$

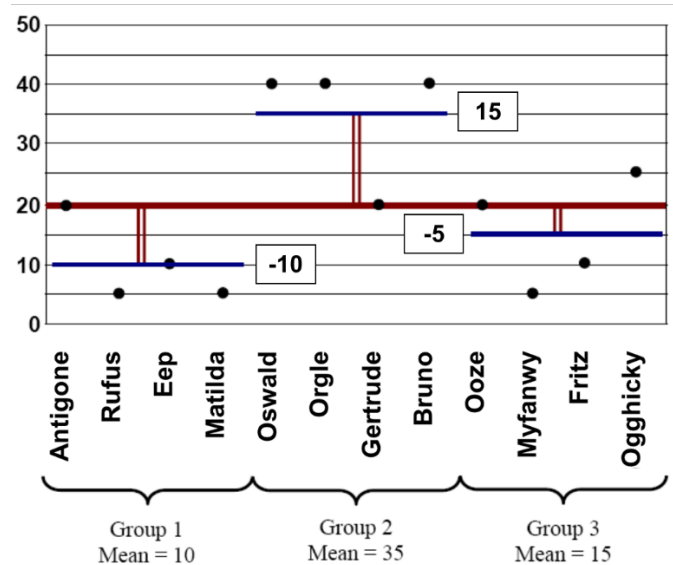
Explained variance is related to the distance between the group means and the grand mean.

If all groups have the same mean, there is no explained variance.

Also called treatment effects.

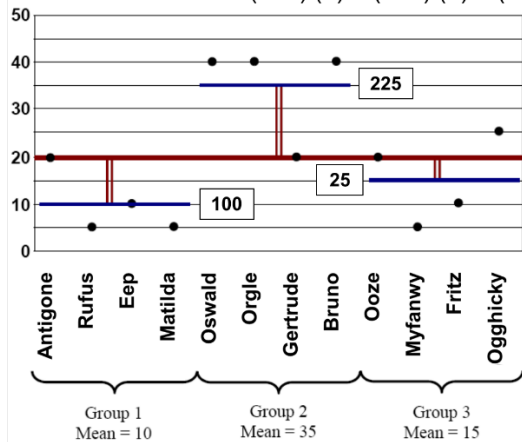
Also called between groups variance.

Group deviations from the grand mean.



Σ (Squared GROUP Deviation) (Group N)

$$= (100)(4) + (225)(4) + (25)(4) = 1400$$



Explained variance

$$SS_{Treatment} = \sum (\bar{y}_{group} - \bar{y}_{Grand})^2 (n_{group})$$

Analysis of Variance (ANOVA)

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained Between groups		p	SS / df	$MS_{Explained} / MS_{Error}$
Error Within groups		$n - p - 1$	SS / df	
Total		$n - 1$		

p = number of contrasts that are free to vary

$p = (\text{number of groups}) - 1$

(Commonly notated as $k - 1$)

Possible comparisons with three groups

	vs.		$\bar{y}_1 - \bar{y}_2 = 5$ Known
	vs.		$\bar{y}_1 - \bar{y}_3 = -20$ Known
	vs.		$\bar{y}_2 - \bar{y}_3 = ?$ NOT FREE

ANOVA

OUTCOME

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	1400	2	700.000	9.000	.007
Within Groups	700	9	77.778		
Total	2100	11			

$$sd = \sqrt{\frac{\Sigma(y - \bar{y})^2}{df}} = 13.817$$

$$sd_{est} = \sqrt{\frac{\Sigma(y - \hat{y})^2}{df}} = \sqrt{77.778} = 8.819$$

What else is this?

$$y - \hat{y} = y - \bar{y}_{group}$$

sd_{est} is the pooled within group sd

Effect Size

Eta-Squared η^2

Analogous to R^2

Like R^2 , η^2 is biased

A less biased estimate.

Omega squared ω^2

Will be smaller than η^2

$$\eta^2 = \frac{SS_{\text{explained}}}{SS_{\text{total}}}$$

$$\omega^2 = \frac{SS_{\text{explained}} - (df_{\text{explained}})(MS_{\text{error}})}{SS_{\text{total}} + MS_{\text{error}}}$$

Planned contrasts

Coding: Create predictor (x) variables with values determined by group membership

Review: 2 groups

y = outcome variable, happiness

Group one (aroma therapy), $x = 1$

Group zero (control), $x = 0$

Group 0 mean $\hat{y}_i = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x_i$ Difference between groups Expected increase for group 1 <i>Interpretation assumes x scored 0 and 1</i>	Regression Expected y when $x = 0$ $\hat{y} = b_0 + b_1x$ Increase in y given a 1 unit increase in x <i>Interpretation assumes x scored 0 and 1</i>
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The following applies regardless of x scoring

$$r = \frac{\text{COV}_{xy}}{sd_x sd_y}$$

$$r_{pb} = \sqrt{\frac{t^2}{t^2 + df}}$$

r = standardized beta

	SS	df	MS	F
Explained (regression)				
Error (residual)				
Total				

Regardless of x scoring

ANOVA tables are the same

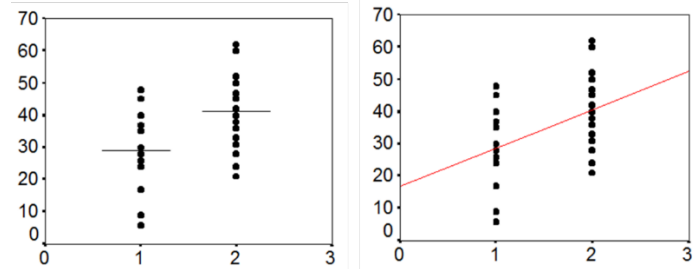
$$\hat{y} = b_0 + b_1x = \text{group mean}$$

	SS	df	MS	F
Explained Between groups				
Error Within groups				
Total				

$$y - \bar{y} = y - \hat{y}$$

In the figure below, x is scored as 1 and 2

Regardless, slope always gives group mean as predicted value.



$$r^2 = R^2 = \eta^2$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{sd_{\bar{x}_1 - \bar{x}_2}} \quad t = \frac{b}{sd_b} \quad t^2 = F = \frac{MS_{\text{explained}}}{MS_{\text{error}}}$$

All p-values will be the same

But, what if we have more than two groups? Create multiple contrasts.

	 control Group1	 Group 2	 Group 3
x_1	-.66	.33	.33
x_2	0	-.5	.5

Create a code assignment chart

$$y = b_0 + b_1x_1 + b_2x_2 + e$$

Each contrast produces an x variable that is entered into a regression equation.

Number of contrasts
=
number of groups
minus one

$$\text{Group 1 : } \hat{y} = b_0 + b_1(-.66) + b_2(0)$$

$$\text{Group 2 : } \hat{y} = b_0 + b_1(.33) + b_2(-.5)$$

$$\text{Group 3 : } \hat{y} = b_0 + b_1(.33) + b_2(.5)$$

Dummy coding:

- Zeros and ones
- Limited use for advanced analyses

Orthogonal contrasts:

- More useful than dummy coding
- *Has several rules. . .*

	 Group 1	 Group 2	 Group 3	
x_1	-.66	.33	.33	$-.66 + .33 + .33 = 0$
x_2	0	-.5	.5	$0 - .5 + .5 = 0$

$$y = b_0 + b_1x_1 + b_2x_2 + e$$

- Values for each contrast must add up to zero.
- Each row must add up to zero.
- For any two contrasts, sum of products must add up to zero.

$$(-.66)(0) + (.33)(-.5) + (.33)(.5) = 0$$

If each contrast has exactly two non-zero values. . .

1. It will compare the two non-zero sets.
2. Any group assigned zero drops out.
3. Interpretation is easiest when the two non-zero values are one unit apart.

$$|(-.66) - (.33)| \approx 1$$

$$|(-.5) - (.5)| = 1$$

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

Expected y when all x variables are 0

“un-weighted” grand mean of y

(mean of the means not weighted by group size)

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

$$b_0 = \frac{\bar{y}_1 + \bar{y}_2 + \bar{y}_3}{3} = \bar{\bar{y}}$$

$\hat{y} = b_0 + b_1x_1 + b_2x_2$ $b_1 = \left(\frac{\bar{y}_2 + \bar{y}_3}{2} \right) - \bar{y}_1$	Increase in y when x_1 increases one unit = How much treated groups exceed the control Hint: to get the correct sign on the beta <i>groups assigned positive code minus groups assigned negative code</i>
$\hat{y} = b_0 + b_1x_1 + b_2x_2$ $b_2 = \bar{y}_3 - \bar{y}_2$	Increase in y when x_2 increases one unit = difference between two treated groups Group 1 drops out: $\hat{y} = b_0 + b_1(-.66) + b_2(0)$ Hint: to get the correct sign on the beta <i>groups assigned positive code minus groups assigned negative code</i>

There are many orthogonal contrast options

4 group example:

	Group A	Group B	Group C	Group D
X1	.5	.5	-.5	-.5
X2	.5	-.5	0	0
X3	0	0	.5	-.5

Omnibus Effects

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

$$\eta^2 = \frac{SS_{\text{explained}}}{SS_{\text{total}}}$$

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained Between groups		p	SS / df	$\frac{MS_{\text{Explained}}}{MS_{\text{Error}}}$
Error Within groups		$n - p - 1$	SS / df	
Total		$n - 1$		

Cohen's d **2 groups**

$$d = \frac{\bar{x}_1 - \bar{x}_2}{sd_{pooled}}$$

$$sd_{pooled} = \sqrt{\frac{SS_1 + SS_2}{df_1 + df_2}}$$

Pooled sd for
2 groups**More than 2
groups**

$$d = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{MS_{error}}}$$

$$sd_{est} = \sqrt{MS_{error}}$$

Pooled within
group sd for
all groups

$$\hat{y} = b_0 + b_1x_1 + b_2x_2$$

$$d = \frac{\left(\frac{\bar{y}_2 + \bar{y}_3}{2}\right) - \bar{y}_1}{\sqrt{MS_{error}}} = \frac{b_1}{\sqrt{MS_{error}}}$$

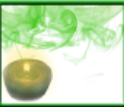
$$d = \frac{\bar{y}_3 - \bar{y}_2}{\sqrt{MS_{error}}} = \frac{b_2}{\sqrt{MS_{error}}}$$

$$t = \frac{b}{sd_{beta}}$$

Post hoc contrasts:

All the possible comparisons

Option: Make all possible comparisons using t -tests

	vs.		$\hat{y}_i = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x_i$ $t = \frac{\bar{y}_1 - \bar{y}_0}{sd_{\bar{y}-\bar{y}}}$
	vs.		$\hat{y}_i = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x_i$ $t = \frac{\bar{y}_1 - \bar{y}_0}{sd_{\bar{y}-\bar{y}}}$
	vs.		$\hat{y}_i = \bar{y}_0 + (\bar{y}_1 - \bar{y}_0)x_i$ $t = \frac{\bar{y}_1 - \bar{y}_0}{sd_{\bar{y}-\bar{y}}}$

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained Between groups		p	SS / df	$\frac{MS_{Explained}}{MS_{Error}}$
Error Within groups		$n - p - 1$	SS / df	
Total		$n - 1$		

$df = \text{error } df$
 $df \neq n - 2$

$t = \frac{\bar{y}_1 - \bar{y}_0}{sd_{\bar{y}-\bar{y}}}$
 Use MS error to calculate $sd_{\bar{y}-\bar{y}}$

2 groups

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{sd_{pooled}^2}{n_1} + \frac{sd_{pooled}^2}{n_2}}$$

Based on pooled sd for 2 groups

More than 2 groups

$$sd_{\bar{y}-\bar{y}} = \sqrt{\frac{MS_{error}}{n_1} + \frac{MS_{error}}{n_2}}$$

Based on pooled within group sd for all groups

family-wise error rate

A sample of 43 people

Asked how much they like 20 random things

Tests for gender differences

	<i>d</i>	<i>p</i>		<i>d</i>	<i>p</i>
Socks	.07	.83	Tomatoes	.34	.31
Cups	-.43	.20	Chalk	.75	.03
Seat belts	.60	.08	Street lights	-.20	.54
Thermostats	.27	.42	Ferrets	.07	.82
Gophers	-.30	.37	Zipper	-.20	.54
Wombats	.11	.73	Pigs	-.59	.08
Goats	-.06	.85	Tape	.08	.81
Sidewalks	-.11	.74	Nails	-.09	.77
Boxes	.42	.21	Broccoli	.07	.82
Magnets	-.28	.31	Bricks	.20	.54

$$\text{possible comparisons} = \frac{k(k-1)}{2}$$

k = number of groups

Fisher's Least Significant Difference Procedure (LSD)

A significant overall F-value gives you license to interpret t-tests for comparisons.

Will not sufficiently protect against family-wise error.

Bonferroni correction

Multiply the obtained p value by the number of pairs tested.

	d	p
Chalk	.75	.03

There were 20 t -tests

corrected p -value is $(.03) (20) = .6$

Advantage

- Easy to use
- Ensures family-wise error is not inflated

Disadvantage

- Often over-corrects

Tukey's Honestly Significant Difference test

Based on studentized range statistic (q)

A linear function of t :

$$q = t\sqrt{2}$$

Tables provide critical values for q based on the number of group means being compared.

Advantage

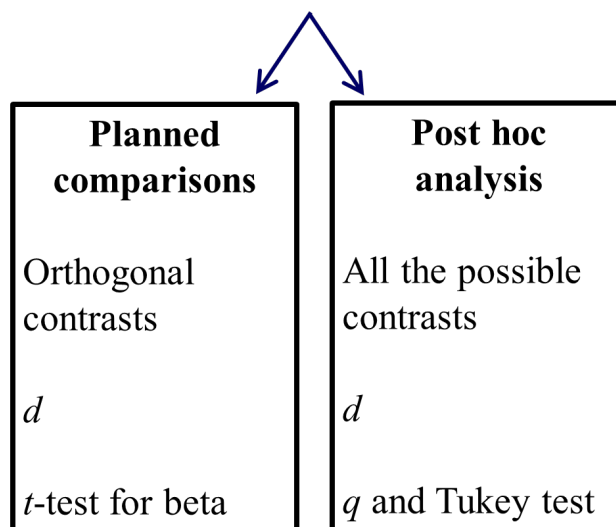
- Ensures family-wise error is not inflated
- Less likely than Bonferroni to over-correct

Disadvantage

- Not convenient for hand calculation

Omnibus Effects

$$\eta^2 = \frac{SS_{\text{explained}}}{SS_{\text{total}}}$$



Research on Couples Therapy

Snyder, D. K., & Wills, R. M. (1989). Behavioral versus insight-oriented marital therapy: Effects on individual and interspousal functioning. *Journal of Consulting and Clinical Psychology*, 57, 39-46.

Types of couple therapy

Behavioral (BCT)
Integrative Behavioral (IBCT)
Cognitive Behavioral (CBCT)
Insight Oriented (IOCT)
Emotionally Focused (EFCT)

Treatment conditions

IOCT: 30 couples
BCT: 29 couples
Waiting list control: 20 couples

Outcome

Global Distress on the MSI

Treatment methods

5 therapists
Each administered both treatments
Twice weekly for 3 months

Descriptives

MSI

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
BCT	29	56.87	11.07	2.06	52.66	61.08	32.45	79.65
IOCT	30	54.06	9.16	1.67	50.64	57.48	32.43	74.33
control	20	64.48	8.67	1.94	60.42	68.54	45.89	78.09
Total	79	57.73	10.52	1.18	55.38	60.09	32.43	79.65

ANOVA

MSI

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	1335.90	2	667.95	6.96	.002
Within Groups	7292.69	76	95.96		
Total	8628.59	78			

Contrast Coefficients

Contrast	group		
	BCT	IOCT	control
1	.33	.33	-.66
2	.5	-.5	0

Contrast Tests

		Value of Contrast	Std. Error	t	df	Sig. (2-tailed)
MSI	Assume equal variances	-5.95	1.67	-3.56	76	.001
		1.40	1.28	1.10	76	.275
	Does not assume equal variances	-5.95	1.55	-3.84	38.01	.000
		1.40	1.33	1.06	54.35	.294

Multiple Comparisons

Dependent Variable: MSI

			Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
(I) group	(J) group					Lower Bound	Upper Bound
TukeyHSD	BCT	IOCT	2.81	2.55	.517	-3.29	8.91
		control	-7.61*	2.85	.025	-14.41	-.80
	IOCT	BCT	-2.81	2.55	.517	-8.91	3.29
		control	-10.42*	2.83	.001	-17.18	-3.66
	control	BCT	7.61*	2.85	.025	.80	14.41
		IOCT	10.42*	2.83	.001	3.66	17.18
LSD	BCT	IOCT	2.81	2.55	.275	-2.27	7.89
		control	-7.61*	2.85	.009	-13.28	-1.94
	IOCT	BCT	-2.81	2.55	.275	-7.89	2.27
		control	-10.42*	2.83	.000	-16.05	-4.78
	control	BCT	7.61*	2.85	.009	1.94	13.28
		IOCT	10.42*	2.83	.000	4.78	16.05
Bonferroni	BCT	IOCT	2.81	2.55	.824	-3.44	9.05
		control	-7.61*	2.85	.028	-14.58	-.64
	IOCT	BCT	-2.81	2.55	.824	-9.05	3.44
		control	-10.42*	2.83	.001	-17.34	-3.49
	control	BCT	7.61*	2.85	.028	.64	14.58
		IOCT	10.42*	2.83	.001	3.49	17.34

*. The mean difference is significant at the .05 level.

Reconstruct the ANOVA table

1. Compute the total sum of squares.
2. Compute the sum of squares between groups.
3. Compute the sum of squares within groups.
4. Compute the degrees of freedom between groups.
5. Compute degrees of freedom within groups.
6. Compute total degrees of freedom.
7. Compute mean square between groups.
8. Compute mean square within groups.

Omnibus question only

Is there a difference between groups? To answer this question, provide the 5 pieces of information.

9. the name of the descriptive statistic
10. the value of the descriptive statistic
11. the name of the inferential statistic
12. the value of the inferential statistic
13. the p-value
14. What is the name of a descriptive statistic giving an effect size that has been corrected for bias?
15. What is the value of the descriptive statistic giving an effect size that has been corrected for bias?

Questions regarding ANOVA table and omnibus effects

16. What is the average distance that people's scores on the outcome variable deviated from the grand mean?
17. What is the average distance that people's scores on the outcome variable deviated from their individual group means? (Calculate a value analogous to the standard error of estimate.)
18. What is the percent of variance that has been explained in this equation?
19. What is the expected score for y if a person is in group 3?
20. If Doug is in Group 1, what is Doug's predicted score on the outcome variable?
21. If Doug actually received a score of 48 on the outcome variable, what is the squared error for Doug's score?
22. According to the results printed above, what is the typical squared error for participants in this study?
23. Was the squared error for Doug higher or lower than the typical squared error for participants in this study?
24. According to the results printed above, what is the sum of squared errors for all participants in this study?

Questions regarding omnibus effects and focused contrasts

Calculate the following:

- b_0
- b_1
- b_2

Assume you have two hypotheses. First, you hypothesize that the two treatment groups will report less marital distress than the control group. Second, you hypothesize that the IOCT group will report less marital distress than the BCT group. Answer the three questions below by providing the five pieces of information.

1. Is there a difference between groups?
2. Do the data support hypothesis 1?
3. Do the data support hypothesis 2?

4. For questions 2 and 3 above, what best describes the size of the effect?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large
5. For questions 2 and 3 above, what would the p-values be if you used the Bonferroni correction?

Now, assume took an entirely different approach to data analysis and that you want to compare all groups to see which groups differ from each other. Based on this approach, answer each of the questions below by providing the five pieces of information.

6. Is there a difference between groups?
 7. Is group 1 different from group 2?
 8. Is group 1 different from group 3?
 9. Is group 2 different from group 3?
10. For questions 7, 8, and 9 above, what best describes the size of the effect?
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large
11. For questions 7, 8, and 9 above, what would the t-values be if you used Fisher's LSD?
 12. For questions 7, 8, and 9 above, what would the p-values be if you used the Bonferroni correction?

Information beyond this point is not needed for this class.

The lecture add-on video includes instructions on how to write a summary of results for analyses that use planned comparisons and post-hoc comparisons, but you will not be asked to write these types of summaries as part of our class.

Summarizing Results from an ANOVA

Planned contrasts

There was a significance difference between groups ($\eta^2 = .15$, $F(2, 76) = 6.96$, $p = .002$). Means and standard deviations for each group are reported in Table 1. Planed contrasts were used to test for group differences. Results from the first contrast indicated that the control group reported more relationship distress than the average of the two treated groups ($d = .92$, $t(76) = 3.56$, $p = .001$). This represents a large effect. The second contrast compared couples receiving behavioral therapy with couples receiving insight oriented therapy, and it was not significant ($d = .29$, $t(76) = 1.10$, $p = .29$).

Table 1.

Group mean differences in relationship distress

	Mean	Standard deviation
Insight oriented couples therapy	54.06	9.16
Behavioral couples therapy	56.87	11.07
Control group	64.48	8.67

NOTE: For planned contrasts, only report contrasts that were truly planned in advance. Do NOT report post-hoc analyses anywhere. Do NOT list any post-hoc analysis results on the table.

Post-hoc comparisons

There was a significance difference between groups ($\eta^2 = .15$, $F(2, 76) = 6.96$, $p = .002$). Means and standard deviations for each group are reported in Table 1. Couples in the control group were significantly more distressed than couples in the insight oriented couple therapy group ($d = 1.07$). This represents a large effect. Control couples were also significantly more distressed than the behavioral couple therapy group ($d = .78$), and this represents a medium effect.

Table 1.

Group mean differences in relationship distress

	Mean	Standard deviation
Insight oriented couples therapy	54.06 _a	9.16
Behavioral couples therapy	56.87 _b	11.07
Control group	64.48 _{ab}	8.67

Means with the same subscript differ from each other at $p < .05$ (Tukey HSD test)

NOTES: For Post-hoc comparisons, do NOT mention non-significant contrasts in the text because listing every single contrast in the text would be difficult to read and redundant with the table. When reporting post-hoc analyses in a table: use subscripts to indicate groups that are significantly different.

Example of table with 4 groups

	Mean	Standard Deviation
Group 1	3.72 _{ab}	14.78
Group 2	5.56 _c	18.53
Group 3	6.43 _a	15.46
Group 4	13.17 _{bc}	19.96

Means with the same subscript differ from each other differ at $p < .05$ (Tukey-HSD test).

In the example above:

- Group 1 differs from Group 2
- Group 1 differs from Group 4
- Group 2 differs from Group 4